



Library for Linear Reference Systems

LibLRS is an Open Source library for railway linear referencing. It supports storing and exchanging position data along railway networks and provides reusable components as well as a standard exchange format for complex infrastructure data.

LibLRS brings together reusable capabilities for working with railway linear referencing data:

- Support for anchors, scales, curves and referencing methods
- Positioning based on anchors and distances along railway lines
- Handling of historical infrastructure inconsistencies and measurement discrepancies
- Geospatial operations to map railway positions to coordinates
- FlatBuffers-based exchange format for sharing data across systems

Experience LibLRS live at InnoTrans 2026:

Hall 7.1C, Booth 530

Explore on GitHub:



<https://github.com/OpenRailAssociation/liblrs>